Status	Finished
Started	Tuesday, 27 October 2026, 9:08 PM
Completed	Tuesday, 27 October 2026, 9:11 PM
Duration	3 mins 24 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**
Complete
Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting decreases the efficiency of selection operations
- b. Sorting helps in finding duplicates quickly
- c. Sorting makes it harder to search for items
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **2**
Complete
Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It sorts data out of the box efficiently
- b. It only works with integer arrays
- c. It requires external libraries
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **3**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Binary Search
- c. Bubble Sort
- d. Linear Search

The correct answer is: Bubble Sort

Question **4**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Binary Search
- b. Linear Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **5**
Complete
Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. sorted()
- c. order()
- d. arrange()

The correct answer is: sorted()

Question **6**
Complete
Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list

Question **7**
Complete
Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Using the divide-and-conquer approach
- b. Being the most efficient sorting algorithm
- c. Bubbling up the largest element to its correct position with each pass
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **8**
Complete
Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Rearranging
- b. Merging
- c. Sorting
- d. Searching

The correct answer is: Sorting

Question **9**
Complete
Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is less efficient than bubble sort
- b. It sorts data using a single pass
- c. It is based on the divide-and-conquer approach
- d. It does not require recursion

The correct answer is: It is based on the divide-and-conquer approach

Question **10**
Complete
Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Extraction
- c. Selection
- d. Distribution

The correct answer is:
Insertion

Question **11**
Complete
Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It complicates the analysis process
- b. It makes the analysis slower
- c. It allows for quicker identification of duplicates
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **12**
Complete
Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 80,150,200,470,40,90,300,400,70,120
- b. 40,70,80,90,120,150,200,300,400,470
- c. 200,470,80,150,40,90,300,400,70,120
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **13**
Complete
Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes data harder to manage
- b. Decreases the efficiency of algorithms
- c. Provides a basis for other algorithms to work efficiently
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**
Complete
Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Sorting the entire list in one pass
- c. Bubbling up the largest element to its correct position
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **15**
Complete
Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Selection sort
- c. Quick sort
- d. Merge sort

The correct answer is:
Merge sort